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MESOZOIC PALEOMAGNETIC RESULTS OF THE TARIM CRATON: TERTIARY RELATIVE MOTION BETWEEN CHINA AND SIBERIA ?

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Abstract. Paleomagnetic poles from two Late Jurassic to Late Cretaceous stratigraphic sections on the northern boundary of the Tarim Craton in western China are concordant with poles of similar age from eastern China. When combined with previously published data, these new results suggest that the major tectonic blocks which comprise China (the Tarim, Yangtze and Sino-Korean cratons) have been arranged in approximately their present relative configuration since Cretaceous times. The averaged Cretaceous paleomagnetic pole for China is significantly different from the averaged Cretaceous pole for Siberia. The difference may reflect significant relative post-Cretaceous motion between the composite China terrane and the Siberian craton. Alternatively, the Cretaceous pole for Siberia may not be reliable.

Introduction

We initiated a paleomagnetic study of Mesozoic rocks from the Tarim craton for two reasons. First, a comparison of the Late Paleozoic apparent polar wander (APW) paths from the Tarim and Siberian cratons had led Li (1988) to propose that Tarim had moved relative to the Siberian craton since Late Permian times. To constrain the timing of this relative motion, Mesozoic paleomagnetic results from the Tarim craton were needed. Second, Courtillot and Besse (1986) pointed out that all Late Jurassic to Early Cretaceous poles for the Sino-Korean and Yangtze cratons come from sites close to or east of the Tanlu fault zone. A test for the post-Cretaceous structural integrity of the major blocks which make up China thus depends critically on results from interior parts of China such as Tarim. To address these questions, we have sampled two Mesozoic sections near Paicheng and Kuche, along the northern margin of Tarim (Figures 1 and 2).

Geological Setting And Sampling

The Paicheng-Kuche region lies on the northern margin of the Tarim craton, where Mesozoic sequences are well developed. One Upper Cretaceous section near Kuche is the Bashenjiqike Formation (K_{2b}), a 180m section of redbeds containing Ziziphocypris simakovi., Rhinocypris cirrita., Cypridea sp., Clinocypris sp. and Mongolianella sp. (Xinjiang Stratigraphic Table, 1981). The Upper Jurassic-Lower Cretaceous section near Paicheng is divided into two formations: the Yageliemu (K_{1y}) and Kelaza (J_{3k}) Formations. The Yageliemu and Kelaza Formations are mainly sandstones and mudstones interbedded with conglomerates. The Yageliemu Formation contains Darwinula fubifomis, Mongolianella palmosa, Pinnocypridea xinjiangensis, Minheella arca, Prolimnocythere earinata, Clinocypris scolia and Lepidotus unguieulatus (Xinjiang Stratigraphic Table, 1981). Deformation in the Paicheng-Kuche area has been active since the Early Cenozoic; this recent folding is probably related to the collision between India and Tibet. We therefore assume that the age of folding is between 45 Ma and present.

Our sampling sites have been located in fine- to medium-grained lithologies. Nine sites, each consisting of six oriented drill cores, were collected from the Paicheng section near the Paicheng Cement Factory. Three sites were located in Upper Jurassic strata and six in Lower Cretaceous strata. Nine sites were collected from the Kuche-Dushanzhi highway section in Upper Cretaceous redbeds, about 60 km north of Kuche. The Paicheng section dips between 50° and 57° to the southeast, while the Kuche section dips between 51° and 72° to the northwest. All cores were oriented by a magnetic compass, supplemented by sun compass readings in most cases.

Paleomagnetic Results

Paicheng

Magnetic measurement and demagnetization procedures were conducted at Stanford University using a cryogenic magnetometer with integrated alternating field (AF) demagnetization coils and a shielded oven. All samples treated by thermal demagnetization were subsequently treated with 5 mT AF before measurement to eliminate viscous effects resulting from exposure to the earth's magnetic field while being transferred from oven to magnetometer. Bulk magnetic susceptibilities were monitored after each heating step.

Intensities of natural remanent magnetization (NRM) of the Yageliemu and Kelaza Formations were typically on the

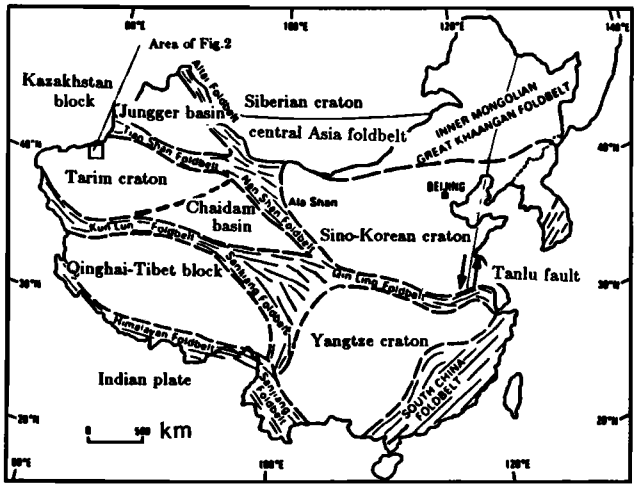


Fig. 1. Generalized tectonic map of China (modified from McElhinny, 1985).

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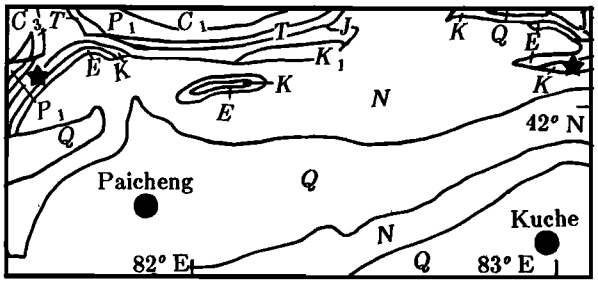


Fig. 2. Generalized geological map of the Paicheng-Kuche area. Q: Quaternary; N: Late Tertiary; E: Early Tertiary; K: Cretaceous; J: Jurassic; T: Triassic; P: Permian; C: Carboniferous. Stars are sampling sections.

order of 10^{-3} A/m. The directions of magnetization for the majority of the specimens do not change after thermal demagnetization to 600° C, suggesting that the characteristic remanent magnetization (ChRM) is probably carried by hematite (Figure 3). Progressive thermal demagnetization at a minimum of 8 heating levels were used to resolve the ChRM components. One component parallel to the present earth's field (PEF) is typically removed in the initial stages of demagnetization (D = 13°, I = 60°, α_{95} = 8°, n = 7 sites, geographic coordinates). Further thermal treatment reveals in most samples only a single high T_{ub} component. The directions of this high T_{ub} ChRM were determined by principal component analysis (Table 1).

ChRM components with southwesterly declination and very shallow inclination from site 154 are quite different from those of the other sites. The magnetizations observed

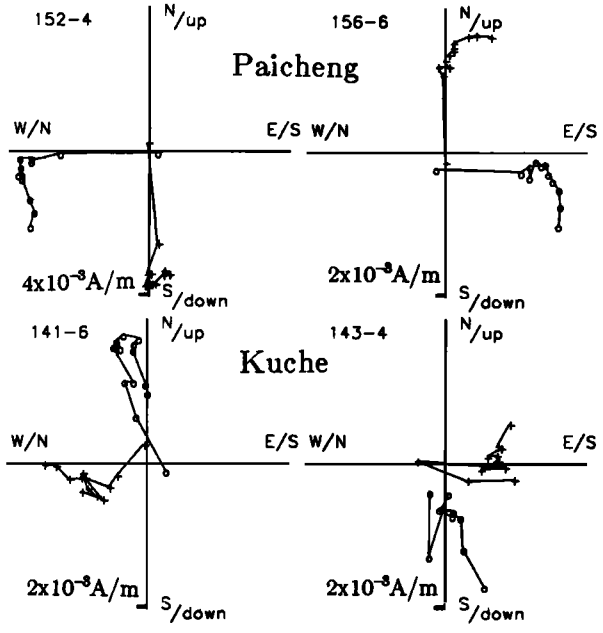


Fig. 3. Orthogonal vector plots of thermal demagnetizations of four normal and reversed polarity specimens from the Paicheng and Kuche sections in geographic coordinates. Plus Symbol refers to the projection on the horizontal plane; open symbol refers to the projection on a vertical, east-west plane. Demagnetization steps plotted for specimen 152-4 are nrm, 100, 200, 300, 500, 550, 600, 650, 670, 680 and 690 ° C; for specimens 156-6 and 141-6, nrm, 100, 200, 300, 400, 500, 550, 600, 625, 650, 675 and 700 ° C; for specimen 143-4, nrm 150, 300, 450, 500, 550, 600, 625, 650, 675 and 695 ° C.

TABLE 1. Paleomagnetic direction

site	n	D_b	I_b	D_a	I_a	κ	α_{95}
Paicheng section(lat.= 41.8N, long. = 82.0E)							
149	6	176	2	191	-40	9	23
150	6	202	6	210	-23	26	14
151	5	192	0	208	-32	58	11
152	6	176	-1	190	-41	75	8
153	4	180	10	186	-29	25	19
154	5	220	-10	232	-15	8	29
155	5	1	12	23	43	26	25
156	3	356	11	20	48	31	22
157	6	9	20	42	44	35	12
$mean_b$	6	2	-7	-	-	52	9
$mean_a$	6	-	-	22	42	62	9
Kuche section(lat.= 41.6N, long. = 83.5E)							
141	6	257	-52	190	-34	45	10
142	2	280	-22	233	-37	-	-
143	6	109	61	26	43	42	10
144	5	118	67	12	44	22	17
146	4	85	61	18	35	24	19
147	4	222	60	311	46	22	20
$mean_b$	4	95	61	-	-	52	13
$mean_a$	4	-	-	16	39	114	9

D_b and I_b or D_a and I_a is site/section mean direction before or after tilt correction; n is number of sample or site in final mean; k is Fisher's precision parameter; α_{95} is radius of 95% confidence limit. mean $_b$ or mean $_a$ is the mean direction of section before or after tilt correction. Sites 142, 147, 150, 153 and 154 are excluded in section mean.

at this site were probably formed during a transitional period of a geomagnetic reversal, since the polarity of the site immediately above is reversed, and the site immediately below is normal. The mean direction of site 154 was therefore excluded from the final average. The inclinations of sites 150 and 153 are also different from those of other sites. The shallow inclinations of site 153 may also be due to transitional fields, while the shallow inclinations of site 150 could result from our inability to demagnetize completely the specimens from this site. We have chosen to exclude these directions from the final average.

Sites 149, 150, 151 and 152 yield reversed polarities, while sites 155, 156 and 157 are normal. As discussed above, sites 153 and 154 probably reflect a transitional polarity. The mean directions of the normal and reversed populations are antipodal at the 95 percent confidence level. Because at least one reversal has been recorded, we think that geomagnetic secular variation may be averaged by the overall mean direction. The observation of dual polarity is in agreement with frequent reversals at the Jurassic-Cretaceous boundary (Harland et al., 1982). We have made no attempt to correlate the observed polarity changes with the time scale, because the ages of magnetization are not yet accurately constrained.

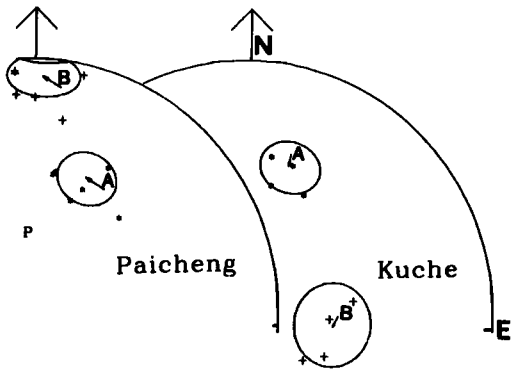


Fig. 4. Equal area stereographic projections of the mean directions of the Paicheng and Kuche sections. + (-) or * is a site mean direction plotted on the lower (upper) hemisphere before or after tilt correction. B or A is the section mean direction before or after tilt corrections (reversed polarity directions are flipped into normal polarity directions). The local present dipole field direction is plotted as a 'P'.

In situ site-mean ChRM directions are discordant with the PEF direction (Figure 4). The magnetizations are not of recent origin, but they approach the PEF direction after correction for tilt (Table 1).

Kuche

In contrast to the Paicheng section, the magnetizations of samples from the Kuche section were not very stable during thermal demagnetization in the high temperature ranges, where intensities and directions changed erratically after 600° C treatment. These changes were associated with increases in magnetic susceptibility, indicating chemical alteration of samples during thermal demagnetization. The direction of the ChRM component in each specimen was determined by principal component analysis over thermal demagnetization steps below 600° C. PEF overprint components were clearly removed in the low temperature range (usually before 300° C). The average low T_{ub} component for seven sites is D = 335°, I = 65°, $\alpha_{95} = 7°$ (in geographic coordinates). After thermal demagnetization, samples from 5 sites show easterly and steep directions in geographic coordinates (Table 1 and Figure 4). Sites 141 and 142 yield antipodal directions.

Discussion of Paleomagnetic Results

Site-mean ChRM directions converted to virtual geomagnetic poles (VGPs) give a mean paleomagnetic pole position for the Paicheng section at long. = 209° E, lat. = 65° N (N = 6 sites, $A_{95} = 9°$), and for the Kuche section at long. = 214° E, lat. = 66° N (N = 4 sites, $A_{95} = 9°$). These two poles, in turn, produce a mean pole at long. = 214° E, lat. = 65° N (N = 10 sites, $K = 66$, $A_{95} = 6°$). Even though the age of the Bashenjiqike (K_{2b}) Formation is different from the ages of the Yageliemu (K_{1y}) and Kelaza (J_{3k}) Formations, the paleomagnetic poles from the two sections converge after tilt corrections are made, providing a positive fold test at the 99 percent confidence level (McFadden and Jones, 1981). This suggests to us that the characteristic magnetization was acquired before the rocks were folded. The small difference in longitude between the poles from the two sections is not statistically significant, but could be due either to apparent polar wander or relative motion between the two sections. Since the magnetizations for the two sections show both polarities and are internally consistent, we consider the results to represent an original magnetization.

Tectonic Interpretations

Available Jurassic and Cretaceous paleomagnetic poles for Siberia and China have been reviewed by Lee et al. (1987) and Courtillot and Besse (1986). Cretaceous poles for the Sino-Korean and Yangtze cratons are essentially identical and are in good agreement with the Cretaceous poles from South Korea (Lee et al., 1987). Eastern China must therefore have been a coherent tectonic unit since Cretaceous times. However, the Cretaceous poles for eastern China are significantly different from those of the Siberian craton. Courtillot and Besse (1986) speculated that this difference could possibly be due to left-lateral displacement along the Tanlu fault, since all the sampling sites for the Sino-Korean and Yangtze cratons are located to the east of this fault. Our new results from the Tarim craton, well west of the Tanlu fault, do not support this interpretation. Cretaceous poles from both sides of the Tanlu fault are clearly consistent (Table 2 and Figure 5).

A selection of Cretaceous paleomagnetic poles available for the Eurasian plate are shown in Table 2. Cretaceous poles for the Siberian craton from several independent studies are not statistically different from each other. A difference does exist between the Cretaceous poles from the Chinese cratons (Sino-Korean, Yangtze and Tarim) and the Siberian craton. It is interesting to note that Cretaceous poles from the Chinese cratons are actually consistent with the Cretaceous pole for the Siberian craton transferred from the North American APWP using rotation poles for closure of the Atlantic (Table 2). One interpretation could be that the measured Cretaceous poles from the Siberian craton are incorrect, and instead the transferred pole should be adopted for the Cretaceous. This pole would then imply that the Chinese and Siberian cratons have been a coherent tectonic unit since the Cretaceous. Alternatively, post-Cretaceous relative motion between the Chinese and Siberian cratons may have occurred due to actual differences reflected in measured Cretaceous poles. The transferred

TABLE 2. Cretaceous paleomagnetic poles of eastern Eurasia

location	age	N	λ_p	ϕ_p	A_{95}	ref.
S. Korean	K	19	68	205	6	Lee et al., 87
NC	K	5	69	200	12	Lin, 84
SC	K	7	76	226	5	Lin, 84
SC	K_3	10	76	173	10	Kent et al., 86
SC	K	?	66	222	4	Hsu, 87
A(mean)	K	5	72	207	7	-
TA	J/K	6	65	209	9	this paper
TA	K_3	4	66	214	9	this paper
B(mean)	K	7	70	208	5	-
NE	K_2	6	74	165	7	Irving and Irving, 82
SI	K	9	68	168	9	Khramov et al., 85
SI	K_1	9	70	166	4	Westphal et al., 86
SI'	K_2	9	69	211	7	Westphal et al., 86

λ_p and ϕ_p: pole position. NC: north China block; SC: south China block; TA: Tarim craton; NE: northern Eurasia; SI: Siberian craton. K is Cretaceous; J/K is Late Jurassic to Early Cretaceous; The additions 1, 2 and 3 stand for Early, Middle and Late respectively. SI is the transferred pole from the North American data. N is number of site or study. A_{95} is radius of the 95 percent confidence limit.

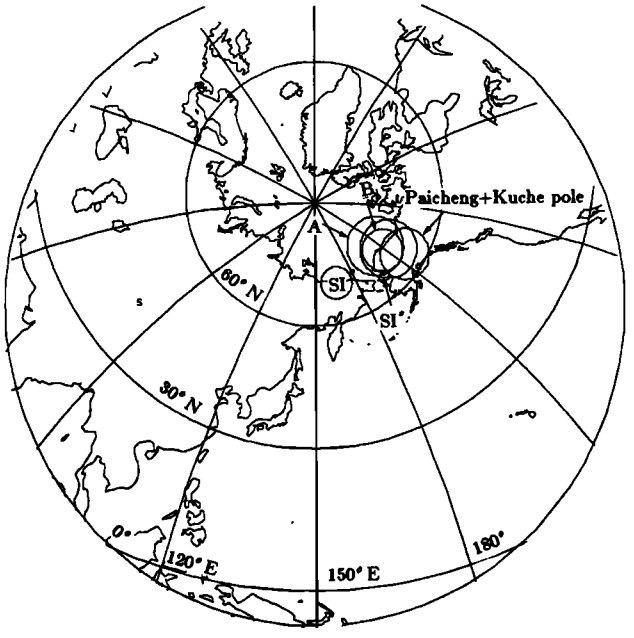


Fig. 5. Oblique polar projection of the Cretaceous poles for the Eurasian plate. See Table 2 for legend.

Cretaceous pole for the Siberian craton is based on marine magnetic anomalies in the Atlantic Ocean, and thus the effects of pre-rift compression and extension are difficult to account for in this type of reconstruction. In such a case, the transferred pole may not necessarily agree with the directly measured pole.

One argument supporting post-Cretaceous relative motion is the similarity in shape between Late Paleozoic APW paths for the Tarim and Siberian cratons (Li, 1988). Paleozoic and Cretaceous poles from Tarim are displaced eastward with respect to their measured Siberian counterparts, implying that Tarim has moved eastward relative to the Siberian craton in post-Cretaceous times (Figure 5). If the Yangtze and Sino-Korean cratons were joined with Tarim after the late Paleozoic but prior to Cretaceous times, the apparent displacement of Cretaceous poles from the amalgamated Chinese cratons may reflect a shift of all three Chinese cratons. The sense of this displacement relative to the Siberian craton can be most easily understood as a consequence of the collision between India and Tibet (Tapponnier et al., 1982).

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